Goutam Barman

Machine Learning | Data Analytics | Full-Stack Development

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GitHub: https://github.com/ipgoutam | Portfolio: https://portfolio-uuwy.onrender.com/

PROFESSIONAL SUMMARY

Aspiring Machine Learning and Full-Stack Developer with hands-on experience building machine learning applications, data analytics dashboards, and full-stack web platforms. Proficient in Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, React, TypeScript, FastAPI, PostgreSQL, and Power BI. Experienced in data preprocessing, exploratory data analysis, predictive modeling, REST API development, database integration, authentication, e-commerce workflows, and responsive web development.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, HTML, CSS
Machine Learning: Scikit-learn, Supervised Learning, Classification, Regression, Predictive Modeling, Data Preprocessing, Feature Engineering, Model Evaluation, EDA
Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
Data Analytics: Power BI, Data Cleaning, Data Visualization, Dashboard Development, Business Intelligence, KPI Analysis
Web Development: React.js, TypeScript, JavaScript, FastAPI, REST APIs, SQLAlchemy, PostgreSQL, HTML5, CSS3, Tailwind CSS, Full-Stack Development
Tools & Platforms: GitHub, VS Code, Jupyter Notebook, Neon PostgreSQL, Render, Cloudinary

PROJECTS

CineMatch – AI Movie Recommendation Platform | Python, Scikit-learn, Pandas, FastAPI, REST APIs, TF-IDF, Cosine Similarity, HTML, CSS, JavaScript, TMDB API

- Developed a full-stack movie recommendation system using content-based filtering, TF-IDF vectorization, and cosine similarity to generate movie recommendations based on content similarity.
- Implemented machine learning workflows and text feature extraction to process movie metadata and identify similarities between movies.
- Built RESTful APIs using FastAPI to integrate the recommendation engine with the frontend and serve movie search and recommendation results.
- Integrated the TMDB API to retrieve movie metadata, descriptions, posters, and other information for enhanced movie discovery.
- Developed a responsive frontend with HTML, CSS, and JavaScript, featuring movie search, recommendations, movie details, and watch list functionality.
- Deployed the application on Render, integrating the frontend, backend, and machine learning components into a live web platform.

Live Demo: https://cinematch-1-wpgm.onrender.com/

G-Mart – Full-Stack E-Commerce Platform | React.js, TypeScript, Python, FastAPI, SQLAlchemy, PostgreSQL, Tailwind CSS, REST APIs, JWT Authentication, Cloudinary

- Developed a full-stack e-commerce platform with separate customer and admin interfaces for managing online shopping operations.
- Built customer-facing features including product browsing, product details, shopping cart, checkout, order history, offers, user profiles, address management, and notifications.
- Developed an admin dashboard to manage products, inventory, categories, brands, homepage content, customers, coupons, orders, payment records, and administrator accounts.
- Built backend REST APIs using FastAPI and SQLAlchemy, with PostgreSQL for persistent storage and frontend-backend integration.

- Implemented user authentication and role-based customer/admin workflows; integrated Cloudinary for product and brand image uploads and management.

Live Demo: https://goat-mart.vercel.app/

Global Airbnb Performance Dashboard | Power BI

- Developed an interactive Power BI dashboard to analyze Airbnb's global market performance across 10 major cities.
- Performed data cleaning, data transformation, and preprocessing to improve data quality and prepare datasets for analysis.
- Created DAX measures and calculated KPIs to track listing performance, host distribution, property types, and market trends.
- Analyzed listing growth, city-wise market share, and property categories, including entire places, hotel rooms, private rooms, and shared rooms.
- Compared average prices across property types to identify pricing differences and understand accommodation market trends.
- Evaluated customer ratings across accuracy, cleanliness, communication, location, and value for money to compare customer satisfaction across cities.
- Designed interactive KPI cards, charts, trend lines, and visualizations to present key insights and support data-driven decision-making.

Live Demo: https://app.powerbi.com/links/fXlwLDJ9wK?ctid=a22bf06c-9316-436a-a631-65a726549c1e&pbi_source=linkShare&bookmarkGuid=329c1d52-d53f-4113-bf53-a6f0cc4ff4c8

EDUCATION

Bachelor of Technology (B.Tech) — Artificial Intelligence & Machine Learning
[Siliguri institute of technology] | [Siliguri, West Bengal] | Expected Graduation: [2028]
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Management Systems, Python Programming, Statistics, Data Analytics, Web Development

ADDITIONAL INFORMATION

Languages: [English, Bengali, Hindi]
Interests: Machine Learning, Full-Stack Development, Data Analytics